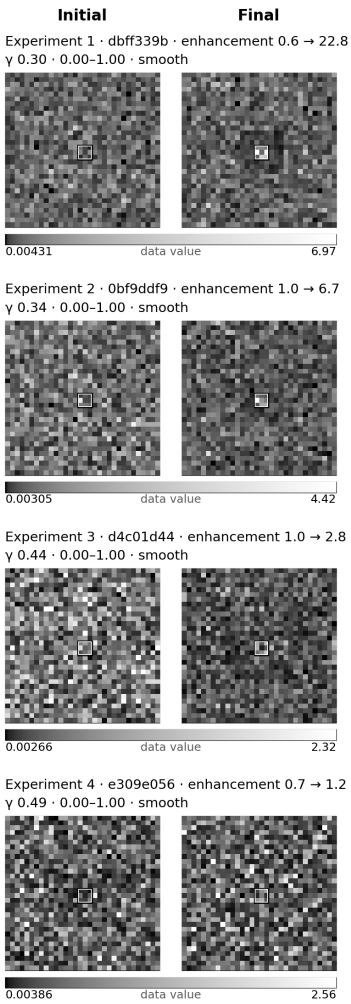
Macropixel size and validated focusing enhancement

Research question: How does DMD macropixel size affect validated focusing when the regions and search budget are fixed?

In this simulation, validated enhancement decreased at each tested increase in macropixel size: 22.79 at size 1 to 1.16 at size 8. Only macropixel size varied; focus/background sizes (3/9), search seed and 1000-iteration budget were shared. The fixed iteration cap constrains fine-grid coverage, and fresh measurement noise differs between runs.

Final tuned heatmaps: Initial and Final panels, experiments in frozen order (macropixel sizes 1, 2, 4, 8).



Macropixel size (px)	Final enhancement	Iterations (count)	Camera shots / measurement
1	22.79	1000	3 search; 5 validation
2	6.68	1000	3 search; 5 validation
4	2.85	1000	3 search; 5 validation
8	1.16	1000	3 search; 5 validation

Verdict and next experiment

All runs used 1000 iterations, 3 camera shots per search measurement and 5 validation shots; wall time and total camera shots were not recorded. Enhancement is the validated focus mean divided by the equally weighted mean of four background-region means.

Finer macropixels gave higher enhancement in this fixed-budget run, but the fine-grid cap and fresh noise limit generalization. Repeat size 1 with focus/background sizes 3/9, 1000 iterations and search seed 43 to test seed sensitivity. Grayscale tuning changes presentation, not measurements; physical print quality was not validated.